FARHEEN RAMZAN

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SHORT BIO

Commonwealth Scholar and PhD Candidate at the University of Sheffield specialising in clinical AI, with doctoral research focused on multimodal deep learning for automated segmentation of myocardial infarction-related scars. Recipient of the **Best Paper Award at AIiH 2025** (Springer, €500), a **UKRI AIRR Compute Award** (10,000 GPU hours on Isambard-AI), and nominated for the **ISW Innovation and Problem Solving Award 2026**. With experience spanning cardiac MRI, ECG, and neuroimaging, and a track record of supervising MSc and BSc researchers, I bring both technical depth and collaborative research experience to clinical AI problems that matter.

EDUCATION

- **University of Sheffield** — *2022 - 2026*
  PhD Computer Science — Sheffield, UK
  - Supervisors: *Prof. Richard H. Clayton* & *Dr. Chen (Cherise) Chen*
  - Thesis: *Towards Automated Segmentation of Cardiac Scars through Multimodal AI*
- **University of Engineering & Technology** — *2017 - 2019*
  MSc Computer Science — CGPA 3.963/4.0 — Lahore, Pakistan
  - Supervisor: *Dr. Muhammad Usman Ghani Khan*
  - Thesis: *Towards a Unified Deep Learning Framework for Alzheimer's Disease Classification using Medical Imaging*
- **University of Engineering & Technology** — *2012 – 2016*
  BSc Computer Science — CGPA 3.847/4.0, **Gold Medal — Ranked 1st in Cohort** — Lahore, Pakistan

HONOURS & AWARDS

- **Nominated: Innovation and Problem Solving Award — ISW Awards 2026** — *May 2026*
  University of Sheffield Careers & Employability Service
  - Nominated by PhD co-supervisor Dr. Chen Chen for independently identifying and solving two critical research challenges: data scarcity (CLAIM framework) and ECG–MRI temporal misalignment (TAFF mechanism), resulting in a **Best Paper Award** at AIiH 2025 and an **oral presentation** at IEEE ISBI 2026.
- **UKRI AI Research Resource (AIRR) Compute Award** — *Jan 2026*
  UK Research and Innovation (UKRI) — Team Award (PI: Dr. Chen Chen)
  - Contributed to the research underpinning a competitively awarded **10,000 GPU hours** on Isambard-AI for *Generative AI–Enabled Digital Twin of the Human Heart with Controllable Scar Patterns*.
- **Best Paper Award** — *Sept 2025*
  International Conference on AI in Healthcare (AIiH 2025) [🌐]
  - Awarded **€500 prize** (sponsored by Springer) for outstanding research contribution at AIiH 2025, Cambridge, UK.
- **Commonwealth PhD Scholarship** — *2022 – 2026*
  UK Foreign, Commonwealth & Development Office [🌐]
  - Prestigious fully funded competitive scholarship to pursue doctoral studies at the University of Sheffield, UK.
- **Gold Medal for Best Overall Performance** — *Nov 2016*
  University of Engineering & Technology, Lahore
  - Awarded Gold Medal for **ranking 1st in cohort** across the entire BSc Computer Science programme.

WORK EXPERIENCE

- **Doctoral Researcher** — *Oct 2022 – Present*
  School of Computer Science, University of Sheffield — Sheffield, UK
  - Conducted original research in multimodal deep learning for cardiac scar segmentation, producing **3 peer-reviewed papers**, **1 oral preprint** (ISBI 2026), and a thesis comprising 4 novel frameworks.
  - Collaborated with NHS clinicians (Nottingham City Hospital) and researchers at the University of Oxford to develop and validate clinical AI pipelines on real patient data.
- **Research Assistant (Part Time)** — *June 2025 – July 2025*
  Chen's Medical Vision Lab, School of Computer Science, University of Sheffield — Sheffield, UK
  - Contributed to *Advancing Cardiac Care through Multi-Modal Data Integration for Precise Scar Mapping*, a **Royal Society funded project** supervised by Dr. Chen Chen.
  - Developed and evaluated multimodal AI pipelines integrating 12-lead ECG signals with cardiac MRI for improved scar segmentation performance.
- **Graduate Teaching Assistant (Part Time)** — *Feb 2023 – May 2024*
  School of Computer Science, University of Sheffield — Sheffield, UK

◦ Supported teaching of COM6103 Team Software Project, COM1003 Java Programming (100+ students/cohort), and COM3524 Bioinspired Computing across four semesters.

- **Lecturer (Junior)** *Sept 2020 – Oct 2022*
  Department of Computer Science, University of Engineering & Technology [🌐] Lahore, Pakistan
  ◦ Designed and delivered lectures in Artificial Intelligence, Human-Computer Interaction, Computer Programming for Data Science, and UX/UI Design to cohorts of **200+ students**.
  ◦ Supervised **5+ undergraduate final year projects and dissertations** across multiple cohorts.
  ◦ Served as **Secretary of the Board of Studies (BOS)**: coordinated governance, faculty recruitment, and curriculum development. Participated in faculty training programmes, seminars, and collaborative course development.

- **Research Officer/Senior Research Officer** *Nov 2016 – Aug 2020*
  Bioinformatics Research Lab, Al-Khawarizmi Institute of Computer Science [🌐] Lahore, Pakistan
  ◦ Conducted research in Machine Learning, Medical Imaging and Brain-Computer Interface. Contributed to software development, academic reporting, and research project management, research papers and grant proposals (national and international).
  ◦ Trained and managed a team of 5–6 researchers across projects, overseeing project delivery and professional development. Served as focal person for academic collaborations.

PUBLICATIONS

C=CONFERENCE, J=JOURNAL

[C.1] **Ramzan, F.**, Kiberu, Y., Jathanna, N., Jabrane, M., Grau, V., Jamil-Copley, S., Clayton, R. H., & Chen, C. (2026). **Seeing Beyond the Image: ECG and Anatomical Knowledge-Guided Myocardial Scar Segmentation from Late Gadolinium-Enhanced Images**. In *IEEE International Symposium on Biomedical Imaging (ISBI) 2026*. *(Oral)*

[C.2] **Ramzan, F.**, Kiberu, Y., Jathanna, N., Jamil-Copley, S., Clayton, R. H., & Chen, C. (2025). **CLAIM: Clinically-Guided LGE Augmentation for Realistic and Diverse Myocardial Scar Synthesis and Segmentation**. In *Artificial Intelligence in Healthcare (AIiH 2025)*, LNCS 16038, pp. 279–292. Springer, Cham. *(Best Paper Award)*

[C.3] **Ramzan, F.**, Chen, C., & Clayton, R. H. (2024). **Improving Myocardial Scar Segmentation with End-to-End and Cascaded CNN using Hybrid Loss and Multi-Modality CMR Imaging**. In *Computing in Cardiology (CinC) 2024*.

[C.4] Rehman, H. A., **Ramzan, F.**, Basharat, Z., Shakeel, M., Khan, M. U. G., & Khan, I. A. (2021). **Genome Scale Phylogenetic Analysis Differentiates SARS-CoV-2 Strains in Pakistani and Indian COVID-19 Patients with and without Travel History**. In *2021 International Conference on Innovative Computing (ICIC)*, pp. 1–9. IEEE.

[C.5] Younis, H., **Ramzan, F.**, Khan, J., & Khan, M. U. G. (2019). **Wheelchair Training Virtual Environment for People with Physical and Cognitive Disabilities**. In *2019 15th International Conference on Emerging Technologies (ICET)*, pp. 1–6. IEEE.

[J.1] Abbas, Q., **Ramzan, F.**, & Khan, M. U. G. (2021). **Acral Melanoma Detection using Dermoscopic Images and Convolutional Neural Networks**. *Visual Computing for Industry, Biomedicine, and Art*, 4(1), 25.

[J.2] Sajjad, M., **Ramzan, F.**, Khan, M. U. G., Rehman, A., Kolivand, M., Fati, S. M., & Bahaj, S. A. (2021). **Deep Convolutional Generative Adversarial Network for Alzheimer's Disease Classification using PET and Synthetic Data Augmentation**. *Microscopy Research and Technique*, 84(12), 3023–3034.

[J.3] Rehman, H. A., **Ramzan, F.**, Basharat, Z., Shakeel, M., Khan, M. U. G., & Khan, I. A. (2021). **Comprehensive Comparative Genomic and Microsatellite Analysis of SARS, MERS, BAT-SARS, and COVID-19 Coronaviruses**. *Journal of Medical Virology*, 93(7), 4382–4391.

[J.4] **Ramzan, F.**, Khan, M. U. G., Iqbal, S., Saba, T., & Rehman, A. (2020). **Volumetric Segmentation of Brain Regions from MRI Scans using 3D Convolutional Neural Networks**. *IEEE Access*, 8, 103697–103709. [108 citations]

[J.5] **Ramzan, F.**, Khan, M. U. G., Rehmat, A., Iqbal, S., Saba, T., Rehman, A., & Mehmood, Z. (2020). **A Deep Learning Approach for Automated Diagnosis and Multi-Class Classification of Alzheimer's Disease Stages using Resting-State fMRI and Residual Neural Networks**. *Journal of Medical Systems*, 44(2), 37. [629 citations]

[J.6] Wasim, M., Sajjad, M., **Ramzan, F.**, Khan, U. G., & Mahmood, W. (2018). **A Review and Classification of Widely Used Offline Brain Datasets**. *International Journal of Advanced Computer Science and Applications*, 9(2).

PRESENTATIONS & TALKS

- ***Seeing Beyond the Image: ECG and Anatomical Knowledge-Guided Myocardial Scar Segmentation from Late Gadolinium-Enhanced Images*** **Oral ×2**
  IEEE ISBI 2026 — London, UK · Workshop on Multimodal AI — London, UK 2025 – 2026
- ***CLAIM: Clinically-guided LGE augmentation for realistic and diverse myocardial scar synthesis and segmentation*** **Oral ×4**
  AIiH 2025 (Best Paper Award) · Insigneo Workshop on Generative AI · Engineering Researcher Symposium · Prof. Iain Wilkinson Seminar — Sheffield, Cambridge, UK 2025

- ***Improving Myocardial Scar Segmentation with Cascaded CNN and Multi-Modality CMR Imaging*** **Poster**
  Computing in Cardiology Conference — Karlsruhe, Germany *Sept 2024*
- ***Using Diffusion Models for Segmentation of Myocardial Scars and Estimating Uncertainty*** **Poster**
  Fickle Heart: UQ, AI and Digital Twins Workshop — Cambridge, UK *June 2024*
- ***Modelling Left Ventricular Scars in Myocardial Infarction through Deep Learning and CMR Imaging*** **Poster ×2**
  Insigneo Showcase (June 2024) · Research Away Day (Jan 2024) — University of Sheffield, UK *2024*
- ***Automated Segmentation of Myocardial Left Ventricular Scar using Deep Learning*** **Poster ×2**
  Insigneo Showcase (July 2023) · COM Research Away Day (June 2023) — University of Sheffield, UK *2023*

Research Projects

- **PhD Thesis: Towards Automated Segmentation of Cardiac Scars through Multimodal AI** *Oct 2022 – July 2026*
  Tools: Python, PyTorch *Domains: Deep Learning, Medical Imaging, Multimodal AI, Diffusion Models*
  - **Multi-sequence CMR segmentation:** Cascaded CNN integrating bSSFP, T2, and LGE with hybrid loss for improved scar delineation.
  - **CLAIM — Clinically-guided scar synthesis:** Conditional diffusion models with coronary territory priors and AHA-17 anatomical constraints to generate physiologically plausible LGE training data. *Best Paper Award, AIiH 2025.*
  - **Seeing Beyond the Image (TAFF):** Multimodal fusion of 12-lead ECG and LGE-CMR via temporal-aware attention to model acquisition asynchrony. *Oral, IEEE ISBI 2026.*
  - **Domain-generalised segmentation:** Modality-masked fusion transformer robust to missing modalities and multi-centre variability. *Manuscript in preparation.*
- **MSc Thesis: Towards a Unified Deep Learning Framework for Alzheimer's Disease Classification** *2017 – 2019*
  Tools: Python, MATLAB, Keras, TensorFlow, Caffe *Domain: Medical Imaging, Deep Learning*
  - Developed a unified framework for multi-class Alzheimer's disease classification and biomarker identification using 4D functional MRI and deep learning.
  - Addressed limitations of binary AD classification by enabling diagnosis across multiple disease stages.
- **Cardiac & Neuroimaging AI** *2016 – 2024*
  Tools: Python, PyTorch, TensorFlow *Domain: Medical Imaging, Deep Learning, BCI*
  - **Brain-Computer Interface (BCI):** Developed EEG-based BCI systems for controlling assistive devices (wheelchair), investigating Motor Imagery, SSVEP, and P300 paradigms.
  - **Brain Region Segmentation:** Automated segmentation of nine brain regions using 3D CNNs with residual learning and dilated convolutions.
  - **Brain Tumor Segmentation:** Deep learning pipeline for automated brain tumor segmentation from MRI with a web-based slice-by-slice visualisation tool.
  - **Liver Lesion Segmentation:** Automated deep learning segmentation of liver lesions in CT images with web-based visualisation.
  - **Alzheimer's Disease (rs-fMRI):** Multi-label classification of six AD stages from resting-state fMRI using brain functional connectivity networks and Stacked Sparse Auto-Encoders.

Student Projects Supervised

- **University of Sheffield** *2025 – 2026*
  Sheffield, UK
  - **Grace D'Abdank Kunicki (BSc, 2025–26):** *Segmentation of myocardial LV scar from LGE-MRI using deep learning* — domain generalization across multi-center datasets.
  - **Meryem Jabrane (MSc, 2025):** *Multimodal ECG and MRI for LV myocardial scar segmentation* — multimodal fusion of temporal ECG and spatial MRI features.
- **University of Engineering & Technology (UET), Lahore** *2019 – 2022*
  Lahore, Pakistan
  - **Anam Anwar (MSc, 2021–22):** *DaT-SPECT based Parkinson's disease and SWEDD classification* — multimodal deep learning with motor and non-motor assessments.
  - **Bushra Tayyaba (MSc, 2021–22):** *Multiclass breast cancer histopathology classification* — transfer learning with stain normalization for improved robustness.
  - **Qaiser Abbas (MSc, 2019–20):** *Detection of Acral Lentiginous Melanoma in dermoscopic images* — deep learning on a South Korean clinical dermoscopy dataset.
  - **BSc Projects (2021–22):** Supervised four BSc projects spanning Parkinson's early onset prediction, gamified learning for autistic children, AI-driven recruitment automation, and university management systems.

Skills

- **Programming Languages:** Python, R, MATLAB, C, C++, C#, Java
- **Deep Learning & ML Frameworks:** PyTorch, TensorFlow, Keras, Caffe, Theano, Weka, Scikit-learn
- **Generative AI:** Diffusion Models (DDPM, DDIM, Latent Diffusion), Stable Diffusion (training & fine-tuning)
- **Medical Image Analysis:** Cardiac MRI (LGE, bSSFP, T2), Brain MRI/fMRI, Chest X-ray, Abdominal CT; Classification, Segmentation, Synthesis
- **Signal Processing:** ECG (NeuroKit, paper-ECG digitisation), EEG (BCI, Motor Imagery, Feature Extraction, Classification)
- **DevOps & Tools:** Git, GitHub, Ubuntu/Linux, LaTeX, Adobe XD, Microsoft Visio
- **Research Skills:** Academic Writing, Grant Proposals, Peer Review, Dataset Curation & Ethical Governance, Project Management

Training & Development

- **Medical Image Computing Summer School (MedICSS)** *July 2024*
  University College London (UCL), London, UK [🌐]
  - Week-long intensive summer school covering state-of-the-art methods in medical image computing and AI.
- **MyOPS++ Scientific Challenge** *July – Aug 2024*
  MICCAI 2024
  - Participated in the MyOPS++ challenge on myocardial pathology segmentation, held in conjunction with MICCAI 2024.
- **MYOSAIQ Scientific Challenge** *June – Aug 2023*
  12th International Conference on Functional Imaging and Modeling of the Heart (FIMH), Lyon, France
  - Participated in an international challenge on myocardial scar segmentation and AI quantification, organised in conjunction with FIMH 2023.

Leadership Experience

- **AI Panelist** *May 2026*
  Sheffield Women in Technology, University of Sheffield [🌐]
  - Invited to represent AI research on a panel alongside academic and industry experts at a Women in Technology meet-up.
  - Shared perspectives on AI progress and interdisciplinary research to an audience of students, researchers, and professionals.
- **Research Team Lead** *July 2019 – Aug 2020*
  Bioinformatics Research Lab, Al-Khawarizmi Institute of Computer Science, UET Lahore [🌐]
  - Led and managed a team of 5–6 researchers across multiple concurrent projects spanning Medical Imaging, Brain-Computer Interface, Bioinformatics, and Drug Discovery.
  - Oversaw project planning, team training, academic reporting, and delivery of national and international grant proposals.
  - Served as focal person for academic collaborations and represented the lab at conferences, seminars, and workshops.
- **Secretary, Board of Studies (BOS)** *Sept 2020 – Oct 2022*
  Department of Computer Science, University of Engineering & Technology, Lahore [🌐]
  - Served as departmental secretary for the Board of Studies, coordinating meetings, agendas, and academic governance activities.
  - Contributed to faculty recruitment processes at both undergraduate and graduate levels.
  - Participated in curriculum development and departmental policy decisions in collaboration with senior faculty.
- **Research Supervisor** *2019 – Present*
  University of Sheffield & University of Engineering & Technology, Lahore
  - Supervised 4 postgraduate (MSc) and 15+ undergraduate (BSc) student research projects across clinical AI, medical imaging, and bioinformatics.
  - Mentored students at the University of Sheffield on cardiac AI projects directly aligned with doctoral research.
  - Provided ongoing guidance on research methodology, scientific writing, and project management to junior researchers.

Volunteer Experience

- **AI Panelist** *May 2026*
  Sheffield Women in Technology, University of Sheffield [🌐]
  - Invited as a panelist at a Women in Technology meet-up to discuss AI research and its applications alongside academics and researchers from the University of Sheffield.
- **Demonstrator** *Nov 2023*
  Medical Physics Outreach Event, University of Sheffield
  - Volunteered as a demonstrator for **50 A-level pupils** at a Medical Physics Outreach Event.

Professional Memberships

- **Student Member** IEEE, Membership ID: 102082497 *Jan 2026 - Present*
- **Member** of Sheffield Women in Computer Science Society *Dec 2022 - Present*
- **Member** of Insigneo Institute for *in-silico* Medicine, University of Sheffield *Nov 2022 - Present*

Certifications

- **Data or Specimens Only Research:** Collaborative Institutional Training Initiative (CITI) Program *Feb 2026*
- **Information Management:** University of Sheffield *Oct 2025*
- **Cyber Security:** University of Sheffield *Oct 2025*
- **Data Protection:** University of Sheffield *Sept 2024*
- **Protecting Research Data:** University of Sheffield *Oct 2023*

References

References available on request.